Electrical Fuse Schedule (Implementation - Lynx Topology)

As-of date: `2026-02-24`

Purpose: define each required fuse by circuit, protected conductor/device, holder/housing method, physical placement, and linked wire gauge assumptions for the approved Phase 1 Lynx architecture with a battery-backed 12V bus.

Related docs:

- Canonical electrical/system baseline: `docs/core/SYSTEMS.md`
- Implementation topology and conductor map: `docs/implementation/ELECTRICAL_overview_diagram.md`
- Battery and trunk recalculation record: `docs/studies/ELECTRICAL_battery_fuse_wire_recalc_2026-02-18.md`
- Decisions/open items tracker: `docs/core/TRACKING.md`
- BOM source of truth: `bom/bom_estimated_items.csv`

Sweep Findings (2026-02-12)

Finding ID	Finding	Impact	Resolution status
`FS-001`	`BB1248120` was modeled in planning docs as `120A` on the `48V` output.	Alternator-recovery math was materially overstated.	Corrected in `bom/bom_estimated_items.csv` row `18` and `docs/core/SYSTEMS.md` (`~1500W`, about `26A` at `57.6V`).
`FS-002`	Fuse schedule listed fuse values but not a complete housing/holder definition for every fuse family.	Install path and procurement freeze were ambiguous.	Resolved below with explicit holder/housing method and location per fuse ID.
`FS-003`	Fuse schedule did not tie each fuse to a conductor gauge plan.	Hard to verify "fuse protects conductor" rule at install time.	Resolved below with gauge mapping and assumptions.

Design Basis

- Topology: `Victron Lynx Distributor M10` (`LYN060102010`) with `4` fused `48V` branches.
- Battery bank assumption: `3x 48V 100Ah` batteries in parallel (`3` separate battery-positive conductors leaving batteries).
- 12V distribution assumption: 12V fuse block used as the shared junction device (main `+` stud = source combine, integrated negative bus/main `-` = return), fed by Orion-Tr Smart `48/12-30` charger and a `12V 100Ah LiFePO4` buffer battery branch.
- Parallel-bank safety rule: use one Class T fuse per battery-positive conductor leaving the battery. A single shared “bank fuse” does not protect the individual battery leads and does not prevent cross-feed faults between parallel batteries.
- Branch devices on Lynx:

1. MultiPlus-II `48/3000`
2. SmartSolar `150/45`
3. Sterling `BB1248120` output
4. Orion-Tr `48/12-30` input

- 12V buffer battery protection and switching baseline: `F-11` (`100A` class) on battery positive within ~`7" of source, then` SW-12V-BATT` manual disconnect for service isolation.
- Phase 1 complexity lock: no automatic low-voltage disconnect (`LVD`) layer in this revision.
- SmartShunt control/power lead note: retain the Victron-supplied fused positive sense/power lead assembly (small-current harness), and avoid substituting lower-voltage automotive fuse components on the `48V` system.
- Sterling rating basis used for this revision: `BB1248120` output ceiling about `1500W` (`~26A` at `57.6V`), with `150A` input fuse and `40A` output fuse guidance.
- Inverter inrush-control baseline: use a manual pre-charge lead across the `48V` main disconnect studs during first energization with `F-02` installed; detailed procedure is in `docs/core/OPERATIONS.md` and procurement is tracked in BOM row `19` .

Manufacturer References Used

- Victron MultiPlus-II `120V` installation page (recommends `125A` DC fuse for `48/3000` ; cable table includes `AWG 1` to `AWG 2/0` by length): `https://www.victronenergy.com/media/pg/MultiPlus-II_120V/en/installation.html`
- Victron SmartSolar MPPT `150/45` installation/spec pages (`50A-63A` battery fuse range, terminal limit `16 mm2` / `AWG 6`): `https://www.victronenergy.com/media/pg/Manual_SmartSolar_MPPT_150-35__150-45/en/installation.html` and `https://www.victronenergy.com/media/pg/Manual_SmartSolar_MPPT_150-35__150-45/en/technical-specifications.html`
- Victron Orion-Tr Smart isolated DC-DC charger product page, manual, and datasheet (`48V` models require external input fuse per manual; `48/12-30` output rating and terminal limits per datasheet): `https://www.victronenergy.com/dc-dc-converters/orion-tr-dc-dc-converters-isolated`
- Victron Lynx Distributor manual (MEGA fuse carrier format): `https://www.victronenergy.com/media/pg/Lynx_Distributor/en/installation.html`
- Victron Orion-Tr Smart isolated DC-DC installation page (external input fuse guidance): `https://www.victronenergy.com/media/pg/Orion-Tr_Smart_DC-DC_Charger_-_Isolated/en/installation.html`
- Victron fuse datasheet (for `48V` systems use `58V` -class fuses): `https://www.victronenergy.com/upload/documents/Datasheet-Fuses-EN.pdf`
- Sterling BB12V->48V charger references (model list includes `BB1248120` , fuse/cable guidance): `https://sterling-power.com/products/12v-to-48v-1500w-battery-to-battery-charger-w-reverse-charging-feature`

- Sterling BB12V->48V charger installation manual PDF (fuse/cable guidance): `https://sterling-power.com/cdn/shop/files/bb1230_1240_1260_12120_122430_241230_241240_241260_241280_242430_124810_1248120_241280_242430_b2b_installation_manual.pdf?v=1739450989`
- Blue Sea Systems guidance (fuse close to source): `https://www.bluesea.com/resources/1437`
- Blue Sea Class T holder-family compatibility (`110A-200A` vs `225A-400A`): `https://www.bluesea.com/support/articles/Fuse_Blocks/1438/Selecting_Class_T_Fuses`

Battery Branch Re-Baseline (2026-02-18)

- This schedule now uses a gated battery-branch baseline of `200A Class T` per battery (`F-01A/B/C`) as a provisional default.
- Re-baseline method and calculations are documented in `docs/studies/ELECTRICAL_battery_fuse_wire_recalc_2026-02-18.md` .
- Final lock gate for `F-01A/B/C` : capture a true `51.2V` battery datasheet/manual with validated current and terminal limits before permanent fuse lock.
- Holder-family constraint: battery branch values below `225A` require the `110A-200A` Class T holder family. Do not pair `<225A` fuse targets with `225A-400A` holder-only hardware.
- If validated battery or terminal limits are lower than current provisional assumptions, move `F-01A/B/C` to `175A` and update BOM row `7` accordingly.

Required Fuse Map (Start-to-Finish, With Housing)

Fuse ID	Circuit (source -> load)	Protected wire/device	Fuse type and voltage class	Amperage	Holder or housing method	Physical location	Planned conductor gauge
`F-01A`	Battery A `+` -> bank positive combine/disconnect input	Battery A positive cable leaving battery	Class T (`>=125VDC`)	`200A` provisional	Blue Sea Class T fuse block (covered stud mount, `110A-200A` family)	Battery compartment, within ~`7"` of Battery A positive post	`2/0 AWG`
`F-01B`	Battery B `+` -> bank positive combine/disconnect input	Battery B positive cable leaving battery	Class T (`>=125VDC`)	`200A` provisional	Blue Sea Class T fuse block (covered stud mount, `110A-200A` family)	Battery compartment, within ~`7"` of Battery B positive post	`2/0 AWG`
`F-01C`	Battery C `+` -> bank positive combine/disconnect input	Battery C positive cable leaving battery	Class T (`>=125VDC`)	`200A` provisional	Blue Sea Class T fuse block (covered stud mount, `110A-200A` family)	Battery compartment, within ~`7"` of Battery C positive post	`2/0 AWG`
`F-02`	Lynx Slot 1 -> MultiPlus `DC+`	Main inverter positive feeder	`MEGA, 58V or 80V`	`125A`	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 1	`2/0 AWG` planned (`AWG 1` minimum on short run)
`F-03`	Lynx Slot 2 -> SmartSolar `BAT+`	MPPT battery-side positive feeder	`MEGA, 58V or 80V`	`60A`	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 2	`6 AWG`
`F-04`	Lynx Slot 3 -> Sterling `BB1248120` output `+`	Sterling output feeder to house `48V` bus	`MEGA, 58V or 80V`	`40A`	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 3	`6 AWG` planned (`10 AWG` minimum per Sterling table)
`F-05`	Lynx Slot 4 -> `48V` auxiliary feeder (Orion branch)	Aux feeder from Lynx to Orion-branch fuse point	`MEGA, 58V or 80V`	`40A`	Integrated Lynx Distributor fuse slot	Lynx Distributor, Slot 4	`6 AWG`
`F-06`	`48V` aux feeder -> Orion `48V` input `+`	Orion input lead (device protection)	Inline MIDI family rated `>=58VDC`	`30A` (`58V` MIDI, lock)	Littelfuse `04980921GXM5` sealed inline MIDI holder on backplate	Electrical cabinet, mounted at source end of Orion input lead	`6 AWG` planned (`8 AWG` minimum per Orion cable table)
`F-07`	Orion `12V` output `+` -> 12V fuse block main `+` stud	Main `12V` feeder from Orion into shared source-combine point	`MEGA, 58V or 80V`	`60A`	Victron MEGA fuse holder (external, non-Lynx)	Electrical cabinet, within ~`7"of Orion12V` output stud	`6 AWG` planned (`8 AWG` minimum per Orion cable table)

F-08	Starter battery + -> Sterling BB1248120 input +	Vehicle-side charger input cable	MEGA/ANL equivalent rated >=32VDC	150A	Sealed engine-bay fuse holder with high-temp cover	Engine bay, within ~`7"` of starter battery positive post	2/0 AWG planned (2 AWG minimum per Sterling table)
F-09A/B/C	PV string + leads -> MPPT PV combiner	Each solar string positive conductor and reverse-current path	gPV string fuse (>=150VDC)	15A each (provisional)	10x38 touch-safe PV fuse holders in weatherproof combiner enclosure	Roof-entry combiner near gland/pass-through	10 AWG PV wire
F-10	12V fuse block branch circuits -> each 12V load	Individual 12V branch conductors and load circuits	ATO/ATC blade fuses (32V class)	Per-circuit	Integrated sockets in generic marine 12V fuse block	12V fuse block in electrical cabinet	Per branch (table below)
F-11	12V buffer battery + -> 12V fuse block main + stud via SW-12V-BATT	Buffer battery source cable and downstream junction fault exposure	Inline MIDI/AMI/ANL family rated >=32VDC	100A class baseline	Sealed inline holder mounted close to battery positive	Within ~`7"` of 12V buffer battery positive post	4 AWG planned
OEM-SHUNT	Lynx positive tap -> SmartShunt positive sense/power lead	SmartShunt electronics lead	Victron OEM inline low-current fuse (factory harness)	OEM value (small-current harness fuse)	Integrated inline holder in supplied harness	Electrical cabinet near Lynx positive tap	OEM harness lead

Notes:

- Victron Orion fuse table is keyed by side voltage (`input or output`). For `48/12-30` : `F-06` tracks the `48V` input side (project lock now `30A` `58V` MIDI based on available holder/fuse family), and `F-07` tracks the `12V` output side (`60A`).

12V Buffer Battery Switching Requirement

- `SW-12V-BATT` is a manual service disconnect in the 12V buffer battery positive path, installed downstream of `F-11` .
- Operational default:

1. `SW-12V-BATT` closed (`NORMAL`) for battery-backed shared-junction operation.
2. `SW-12V-BATT` open (`SERVICE`) for battery isolation or Orion-only validation mode.

- Service-path note: the Orion `12V` output remains connected to the fuse-block main `+` stud through `F-07` even when `SW-12V-BATT` is open.
- Maintenance-path note: when `SW-12V-BATT` is closed, Orion output can maintain/charge the buffer battery through the shared fuse-block main `+` stud path.
- Do not solder-splice high-current source conductors; use crimped lugs on rated studs/junction hardware.
- Keep always-on safety branch (`12V-05`) unswitched at branch level; isolation is handled at source path controls.

12V Branch Fuse + Gauge Assignments (Initial Build)

Circuit	Planned load	Fuse	Planned wire gauge	Notes
12V-01	Starlink DC power supply	10A	14 AWG duplex	Remote-work load path
12V-02	Fridge (35-50qt)	15A	14 AWG duplex	Current estimate pass flags this as a voltage-drop warning at ~12 ft; hold <=8 ft or move to 12 AWG
12V-03	Diesel heater electrical	15A	14 AWG duplex	Startup surge expected; estimate pass flags tight margin at ~8 ft
12V-04	Water pump	10A	14 AWG duplex	Intermittent duty
12V-05	CO + propane detector	3A	18/2	Always-on safety load
12V-06	Factory LED lights + dimmer (Hiatus pre-installed)	5A	18/2	Hiatus factory branch; separate from planned ambient/Govee strip branch
12V-07	Cerbo GX power feed (assumed via 12V panel)	3A	18/2	Assumption until final harness lock
12V-08	USB PD station branch (office zone)	20A	12 AWG duplex	Keep <=5 ft (estimate-pass assumption) or upsize
12V-09	USB PD station branch (galley zone)	15A	14 AWG duplex	Current estimate pass assumes ~8 ft and ~8A expected load; move to 12 AWG if sustained current rises
12V-10	Maxxair fan (Hiatus pre-installed)	10A	14 AWG duplex	Hiatus pre-installed roof fan branch; verify installed harness and branch labeling at audit

12V-11	DC ambient/cabinet LED strips (planned Govee)	5A	18/2	Planned non-factory lighting branch; keep DC-fed to avoid inverter-on requirement for lighting-only use

F-11 Validation (2026-02-22)

- Scope validated: `C-19A` (12V buffer battery positive path), including `F-11`, `SW-12V-BATT`, and planned `4 AWG` conductor.
- Documented branch-fuse envelope from `12V-01` through `12V-11`: `111A` theoretical simultaneous maximum (`10+15+15+10+3+5+3+20+15+10+5`).
- Modeled practical load envelope from `bom/load_model_wh.csv` is materially lower (`~300W` average, about `25A @ 12V`), but branch-level surge/transient overlap remains possible.
- Conductor check for `4 AWG`, `2.5 ft` one-way (`5 ft` loop) using this doc resistance basis (`0.0002485 ohm/ft`):

1. `50A -> 0.062V` drop (`0.52% @ 12V`)
2. `80A -> 0.099V` drop (`0.83% @ 12V`)
3. `100A -> 0.124V` drop (`1.04% @ 12V`)

- Validation result: keep `F-11` at `100A` class for the current architecture. This preserves margin above the modeled `50A` design basis and near/at the branch-fuse aggregate envelope while staying aligned with the `4 AWG` short-run drop target.
- Hard lock conditions for this `100A` choice:

1. Buffer-battery positive path remains `4 AWG` and approximately `2.5 ft` one-way as documented.
2. `SW-12V-BATT` continuous DC rating is `>=100A` at the installed system voltage class.
3. The selected `12V 100Ah` battery/BMS continuous discharge rating is `>=100A`.

- Down-select gate: if final battery/BMS continuous rating is below `100A`, reduce `F-11` to the nearest compliant class (typically `80A`) and synchronize spare stock row `105`.

Length-Validation Sync (2026-02-18)

- `docs/implementation/ELECTRICAL_overview_diagram.md` now carries one-way estimated lengths and voltage-drop screening for `C-01` through `C-37`.
- Lynx Orion-feeder branch fuse (`F-05`) is locked to `40A MEGA` for BOM sync in this pass.
- USB branch baseline is synchronized across docs/BOM: office branch `12 AWG` (`C-34`), galley branch `14 AWG` (`C-35`).

Holder Ecosystem Standard (Procurement Guidance)

Fuse family	Preferred holder style	Why
Class T (`F-01A/B/C`)	Covered stud-mounted marine block (`110A-200A` family for current baseline)	High interrupt capacity and robust battery-compartment mounting; holder family must match chosen Class T range
MEGA (`58V/80V`) in Lynx (`F-02` to `F-05`)	Lynx integrated holders only	Eliminates separate branch fuse block and keeps topology locked
Inline `48V` Orion input (`F-06`)	Littelfuse `04980921GXM5` sealed inline MIDI holder on interior backplate	Keeps source-end protection near branch takeoff with confirmed `58V` holder availability
Orion output (`F-07`) `60A MEGA`	Victron external MEGA holder at Orion output	Standardizes `60A` spare pool with MPPT fuse family
12V buffer battery main (`F-11`)	Sealed MIDI/AMI/ANL holder near battery	Protects shared 12V source-junction path from battery fault current
12V battery disconnect (`SW-12V-BATT`)	Sealed rotary DC switch near distribution cabinet access	Provides simple service isolation without adding LVD complexity
Engine-bay Sterling input (`F-08`)	Sealed MEGA/ANL holder, high-temp/engine-bay suitable	Environmental robustness and source-proximate protection
PV string fuses (`F-09`)	DIN-rail `10x38` gPV holders in weatherproof combiner	Proper PV-voltage insulation and serviceable string isolation
12V branches (`F-10`)	Integrated blade sockets in marine fuse block	Fast field service and clear branch labeling
SmartShunt positive lead (factory harness)	Keep Victron-supplied inline fused lead assembly	Preserves vendor-qualified low-current protection path for shunt electronics

Spare Fuse Inventory

Fuse type	Installed qty	Spare qty to carry	Notes
Class T `200A` (provisional installed)	`3`	`3`	One spare per installed battery fuse while `200A` baseline is active
Class T `175A` (alternate pending final battery datasheet lock)	`0`	`3` optional	Optional non-regret alternate set if budget allows; install only if final lock gate requires lower value

MEGA 125A (58V/80V)	1	4	MultiPlus branch; BOM lock is x5 total
MEGA 60A (58V/80V)	2	4	Shared pool for MPPT (F-03) + Orion output (F-07); target 6 total
MEGA 40A (58V/80V)	2	4	Shared Sterling (F-04) + Orion feeder (F-05) family; BOM lock is x6 total
Orion input fuse (30A MIDI, 58V)	1	3	BOM row 133 lock is x4 total (1 installed + 3 spares)
12V buffer battery main fuse (100A class)	1	3	Spare pack basis is BOM row 105 (BOJACK 100A ANL 3-count)
Sterling input fuse 150A (32V+)	1	1	Vehicle-side charger input
PV string fuse 15A gPV	3	3	One spare per string
SmartShunt OEM harness fuse	1	1	Keep OEM-equivalent spare if the fuse is field-replaceable on final harness
ATO/ATC branch fuses	variable	2 each used value	Keep mixed kit onboard

BOM Row Mapping

Fuse scope	BOM row(s)
Main battery Class T protection (F-01A/F-01B/F-01C) + Class T spares	bom/bom_estimated_items.csv row 7
Lynx branch MEGA fuses (F-02 to F-05) installed + spare set	bom/bom_estimated_items.csv row 10
Orion/Sterling installed fuse-holder hardware (F-06, F-07, F-08)	bom/bom_estimated_items.csv row 11
Orion input fuses (F-06)	bom/bom_estimated_items.csv row 133
12V buffer battery (B12)	bom/bom_estimated_items.csv row 21
12V buffer battery main fuse + holder (F-11)	bom/bom_estimated_items.csv row 125
12V battery disconnect (SW-12V-BATT)	bom/bom_estimated_items.csv row 124
12V branch panel and blade fuses (F-10)	bom/bom_estimated_items.csv row 16
SmartShunt OEM fused lead (OEM-SHUNT)	bom/bom_estimated_items.csv row 23 (included with SmartShunt kit)
High-current spare fuse kit (non-Lynx spares, if retained)	bom/bom_estimated_items.csv row 105
PV string fuses + holder (F-09A/B/C) and spares	bom/bom_estimated_items.csv row 106

Related non-fuse accessory for energization safety:

- Inverter pre-charge lead accessory (`100-150 ohm` resistor lead used across main disconnect studs): `bom/bom_estimated_items.csv` row `19` .

Lynx MEGA Procurement Lock (BOM Sync)

- Updated row `10` fuse-family lock: `125A x5` , `60A x6` (shared `F-03` + `F-07` pool), `40A x6` (shared `F-04` + `F-05` pool).
- Updated row `10` estimate should be kept synchronized with the final `40A x6` cart quantity.

Assumptions and Open Items

1. Wire sizing above assumes copper conductors, enclosed vehicle routing, and the one-way run-length estimate set in `docs/implementation/ELECTRICAL_overview_diagram.md` (`2026-02-18` pass).
2. `F-09A/B/C` remains provisional at `15A` pending final solar module datasheet max-series-fuse confirmation.
3. Final SKU lock is still required for inline holders used by `F-08` and `F-11` , plus the external Victron MEGA holder for `F-07` and `SW-12V-BATT` .
4. If Orion run lengths exceed the short/medium assumption, keep `F-06` / `F-07` and upsize conductors before energizing.
5. Do not use `32V` automotive MEGA fuses on `48V` house circuits; use `58V / 80V` DC-rated fuses only.
6. Keep the SmartShunt fused lead as an OEM harness item unless an equivalent voltage-rated replacement is fully validated.
7. `SW-12V-BATT` is manual-only in Phase 1; no automatic LVD behavior is assumed in this schedule.
8. Battery listing data (`<=200A` current limit) is currently treated as provisional because the provided screenshot includes `14.6V/14.2V` values that are not a `51.2V` profile; final lock requires validated `51.2V` battery documentation.
9. If final battery current or terminal limits are lower than current assumptions, down-select `F-01A/B/C` to `175A` and keep fuse-to-conductor coordination synchronized with `docs/implementation/ELECTRICAL_overview_diagram.md` and `bom/bom_estimated_items.csv` .
10. Orion fuse selection is side-voltage based per Victron table (`input or output`), not just converter model number. Keep `F-06` and `F-07` aligned to `48V` input / `12V` output sides.

11. Final lock for `F-11` requires explicit 12V buffer battery/BMS continuous discharge-current confirmation; if `<100A`, down-select `F-11` and `SW-12V-BATT`/holder family accordingly.